

V • The 20-Scale Map

The same Helmholtz equation governs all 20 scales. Coupling constants $k(\sigma)$ are derived via the Zoom Operator Λ from the Calabi-Yau moduli geometry — they are not free parameters.

- σ 0 Planck / quantum foam — $\hbar_{\text{geo}}, \tau_m \sim 10^{-43}$ s
- σ 1–2 Nuclear / atomic — ion channel gating, $\tau_m \sim 10^{-23}$ s
- σ 3 Molecular — protein folding, receptor binding, $\tau_m \sim$ ms
- σ 4 Synaptic — vesicle release, LTP/LTD, $\tau_m \sim$ ms–s
- σ 5 Cellular — single neuron EM field, spike patterns, $\tau_m \sim$ s
- σ 6 Local circuit — cortical column, gamma oscillation, $\tau_m \sim$ s–min
- σ 7 Whole brain — global somatic field, emotional state, $\tau_m \sim$ min–hr
- σ 8 Dyadic — two-person field, rapport, therapy session, $\tau_m \sim$ hr
- σ 9 Group / swarm — collective field, team coordination, $\tau_m \sim$ hr–days
- σ 10 Community — social trust network, $\text{SQ} \equiv \lambda_1(\hat{L}_{ij}), \tau_m \sim$ yr
- σ 11 Regional — dialect spread, cultural identity, $\tau_m \sim$ decade
- σ 12 National — institutional field, law, $\tau_m \sim$ decade
- σ 13 Civilisational — economic attractor landscape, $\tau_m \sim$ century
- σ 14 Species — evolutionary field, $\tau_m \sim$ millennium
- σ 15–17 Geological — tectonic soma, rock strata memory, $\tau_m \sim 10^3$ – 10^6 yr
- σ 18 Galactic — astrophysical field, $\tau_m \sim 10^8$ yr
- σ 19 Observable universe — cosmological, $\Lambda \equiv \langle \text{tr } \Phi \rangle_0, \tau_m \sim 10^{10}$ yr

Geological memory ($\sigma = 15$ – 17): fault zone geometry encodes past ruptures as Hopfield patterns. Gutenberg-Richter $b \approx 1$ derived from universality class.

VI • Key Identifications Across All Domains

- **SHO** $\equiv$ **Green's function** — the “vibrating string” IS the field's impulse response; derivation, not postulate
- **Nash equilibrium** $\equiv$ **Hopfield minimum** — $\text{Nash}(G) = \arg \min H(\mathbf{e})$; game theory = attractor dynamics
- **Rights** $\equiv$ **topological invariants** — stable under continuous deformation of legal landscape; cannot be removed without topological transition
- **Rule of law** $\equiv$ **ergodicity** — $\bar{f}_{\text{time}} = \bar{f}_{\text{ensemble}}$; power protects some agents from visiting accountability

regions

- **Rapport** $\equiv$ **Huygens frequency locking** — $\dot{\phi}_1 - \dot{\phi}_2 \rightarrow 0$; coupling must exceed locking threshold $\kappa_{\min}$
- **Social Intelligence SQ** $\equiv$ **spectral gap** — $\text{SQ} \equiv \lambda_1(\hat{L}_{ij})$; measures synchronisation rate
- **Market crash** $\equiv$ **somatic phase transition** — $T \rightarrow T_c^+$, $\xi \rightarrow \infty$; systemic risk = proximity to critical point
- **Gutenberg-Richter** $b \approx 1 \equiv$ **universality class** — derived, not empirically fitted
- **Music** $\equiv$ **field perturbation** — tempo = Arnold tongue driving; timbre = field direction; resolution = attractor basin entry
- **Somatic therapy efficiency** — $\eta_{\text{som}}/\eta_{\text{cog}} \approx \kappa_{\text{EC}}^{-1}$; bypasses decoupled EC junction
- **Optimal regulation** $\equiv$ **WKB problem** — minimum intervention strength = barrier height in energy landscape
- **Geologic memory** $\equiv$ **Hopfield pattern** — fault geometry encodes past ruptures; lowers energy barrier for future rupture

VII • Clinical Framework

ASC operator — high β , narrow Arnold tongue, deep stable attractors. Deep domain engagement, costly state transitions, narrow synchronisation bandwidth. Not a disorder — a field architecture with different costs and affordances.

CPTSD operator — non-ergodic field (trauma well dominates landscape), $\kappa_{\text{EC}} \ll 1$ (somatic-cognitive decoupling). Cannot “think out” of trauma: barrier too steep for classical gradient descent. Requires somatic entry or WKB tunnelling.

ADHD operator — high field temperature T , flat landscape (all barriers $\sim D$), rapid transitions between attractors. The same high T that causes dysregulation also enables escape from trauma wells (Anti-Freeze mechanism).

Co-occurrence ($\text{ASC} \cap \text{CPTSD}$): high $\beta \Rightarrow$ deeper trauma wells on adverse input. Not coincidence — structural consequence of the field architecture.

Therapeutic intervention = **optimal control** on field trajectory. Find $J_{\text{therapy}}(x, t)$ that moves field from trauma well to healthy attractor while staying within window of tolerance.

Somatic injection directly perturbs Φ without translation through decoupled EC junction. For CPTSD with $\kappa_{\text{EC}} \ll 1$: somatic $\gg$ cognitive efficiency.

God-Knob = field temperature parameter. SSRIs raise T by serotonergic coupling. Titrated trauma processing = temperature regulation within the therapeutic window.